इसरो isro

BHARATIYA ANTARIKSH HACKATHON 2025

Powered by H2S HACK2SKILL

Team Name : Space Gladiators

Team Leader Name : Shendre Samarth Manish

Problem Statement : AI-based Help Bot for Information Retrieval from a Knowledge Graph Based on Static/Dynamic Web Portal Content

Team Members

Team Leader:

Name: Shendre Samarth Manish
College: Sinhgad College Of Engineering,Pune

Team Member-1:

Name: Shivam Goraksha Dahifale
College: Sinhgad College Of Engineering,Pune

Team Member-2:

Name: Saee Bandal
College: Sinhgad College Of Engineering,Pune

Team Member-3:

Name: Shreya Balakrishna Ghodake
College: Sinhgad College Of Engineering,Pune

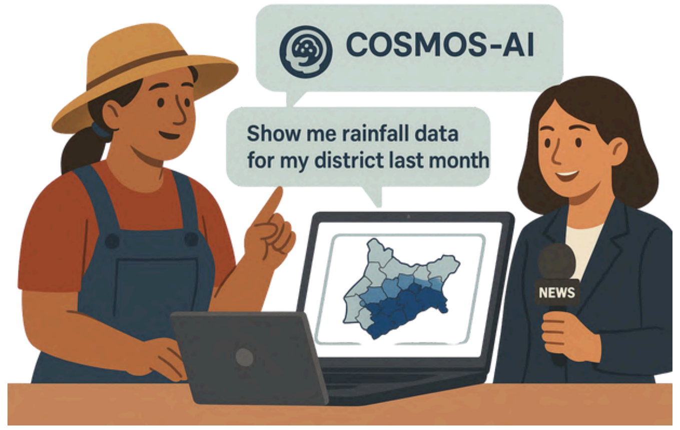
Brief about the Idea:

COSMOS-AI:
Cognitive Observatory for Space & Meteorological Operations Support

COSMOS-AI is an intelligent virtual assistant that transforms how users interact with ISRO's MOSDAC portal by converting complex satellite data and documentation into conversational experiences.

Releveance

Farmer : Show me rainfall data for my district last month" - gets instant maps instead of navigating 15+ menu layers



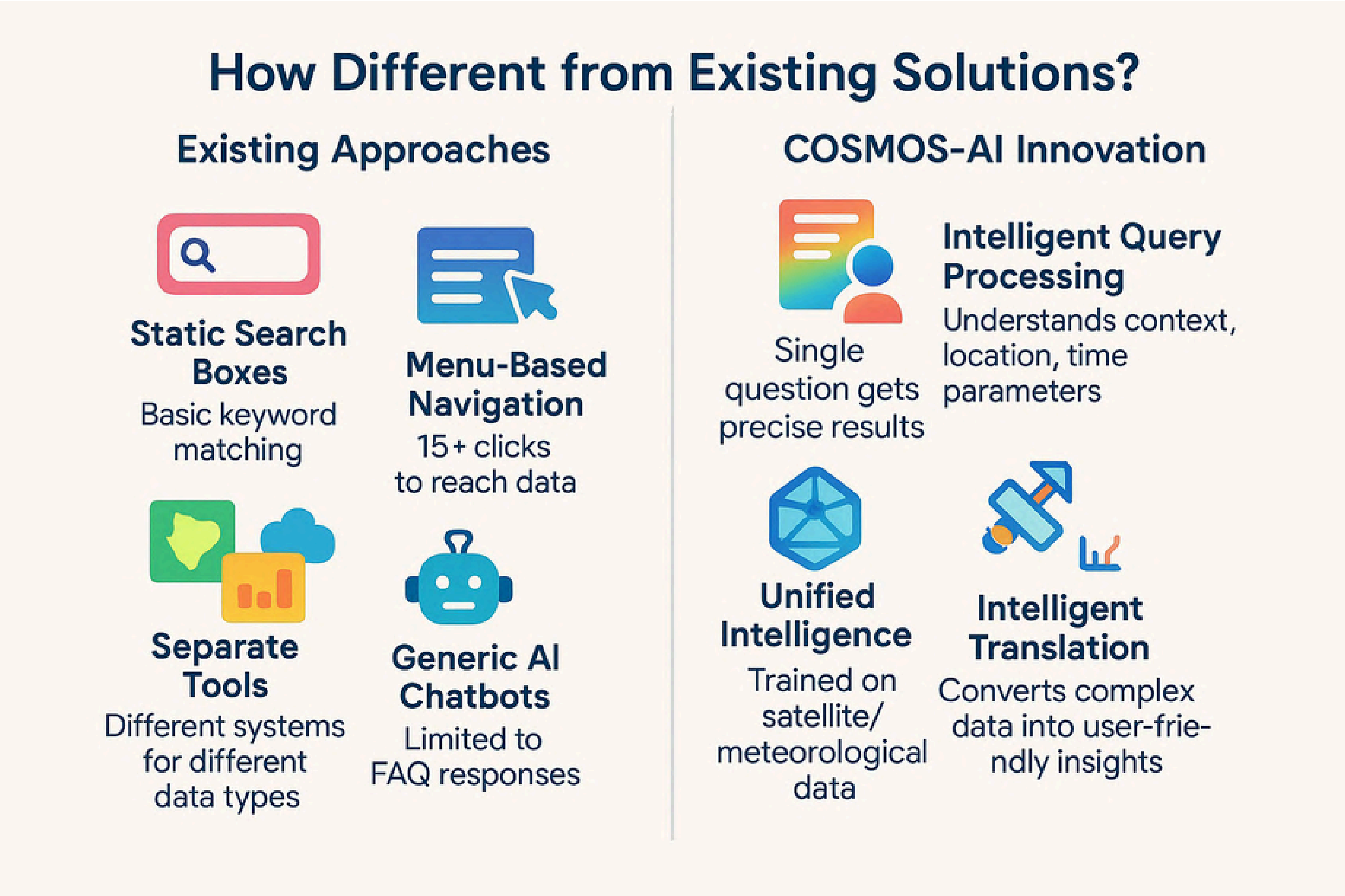
News Report :Need latest satellite images of flooding in Assam" - instant access during breaking news coverage

📋 Overview:

1. Intelligent Data Navigator for MOSDAC content
2. Hybrid AI Architecture (Knowledge Graph + RAG)
3. Multi-Modal Intelligence processing
4. Geospatial Query Engine with location-time capabilities

Opportunity should be able to explain the following:

Powered by H2S



🎯 How It Solves the Problem

MOSDAC contains 10+ years of satellite data across 50+ categories, but users spend 20-30 minutes finding specific information due to complex navigation and technical barriers.

1. Intelligent Data Discovery: Proactive suggestions based on query patterns
2. Context-Aware Processing: Intelligent clarification system for ambiguous queries
3. Real-Time Data Integration: Live satellite feed integration with historical data
4. Multi-Stakeholder Accessibility: Adaptive response formatting based on user profile

⭐ Unique Selling Propositions (USP)

Hybrid AI Engine

Combines Knowledge Graph + RAG for accurate and contextual responses.

Geo-Temporal Query Support

Understands location + time queries like "rainfall in Assam last month".

Conversational Data Access

Turns complex data into simple chat — no need for 15+ menu clicks.

Multi-Format Intelligence

Reads PDFs, DOCX, XLSX, ARIA-tagged HTML in one unified system.

Adaptive Responses by User Type

Scientist, farmer, or journalist — gets info in their preferred format.

Real-Time Satellite Alerts

Integrates live feeds for disasters like floods, fires, cyclones.

Proactive Query Clarification

Smart follow-ups for vague questions — boosts user experience.

Multilingual Support

Works in English and Hindi to reach both experts and rural users.

List of features offered by the solution

Powered by H2S

Natural Language Query Support
Users can ask questions like “satellite images for Kerala flood” in plain English or Hindi.

Multilingual Support (EN + HI)
Allows users from diverse backgrounds to interact effortlessly.

Hybrid AI Processing
Uses both Knowledge Graph and Retrieval-Augmented Generation (RAG) for smart, contextual answers.

User-Adaptive Responses
Customizes answers depending on whether the user is a farmer, scientist, or journalist.

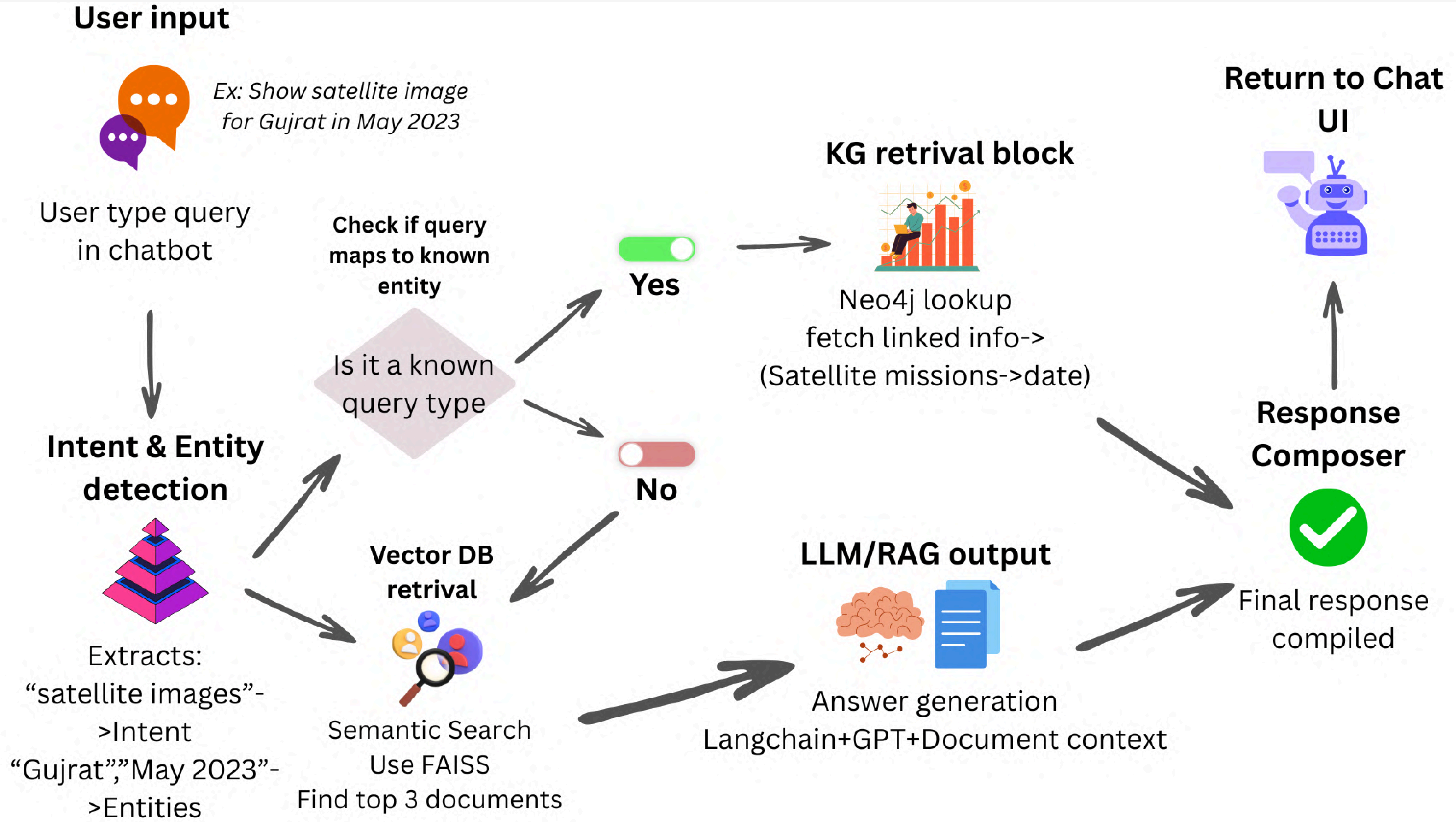
Multi-format Data Ingestion
Crawls and understands data from PDFs, DOCX, XLSX, FAQs, and ARIA-tagged web content.

Live Satellite Feed Integration
Pulls in real-time data and overlays it with historical maps for dynamic insights.

Geo-Temporal Query Engine
Understands queries based on location + time (e.g., “rainfall in Gujarat last August”).

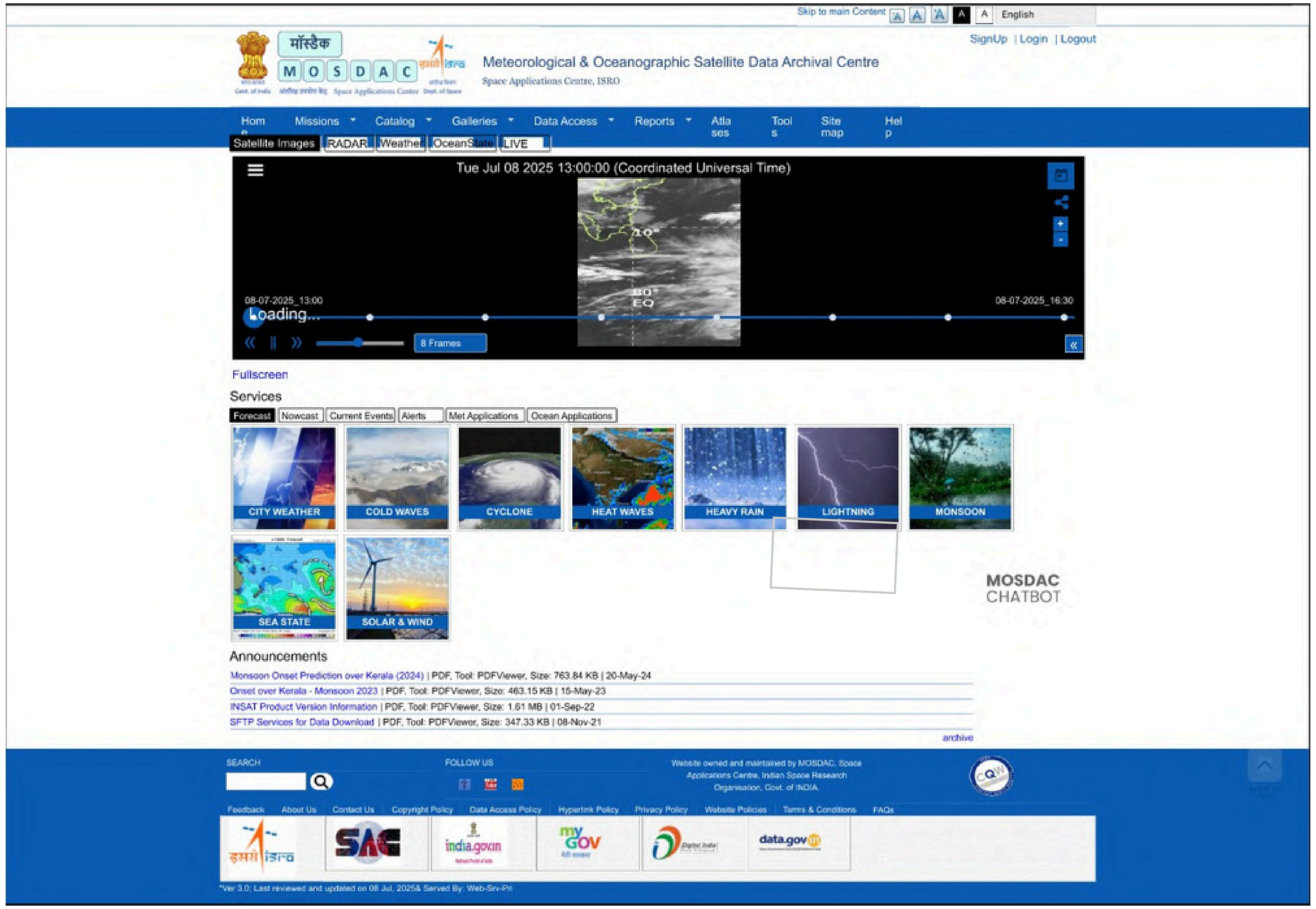
Conversational UI Chatbot
Smart, multi-turn conversations with context memory and follow-up clarification.

Process flow diagram

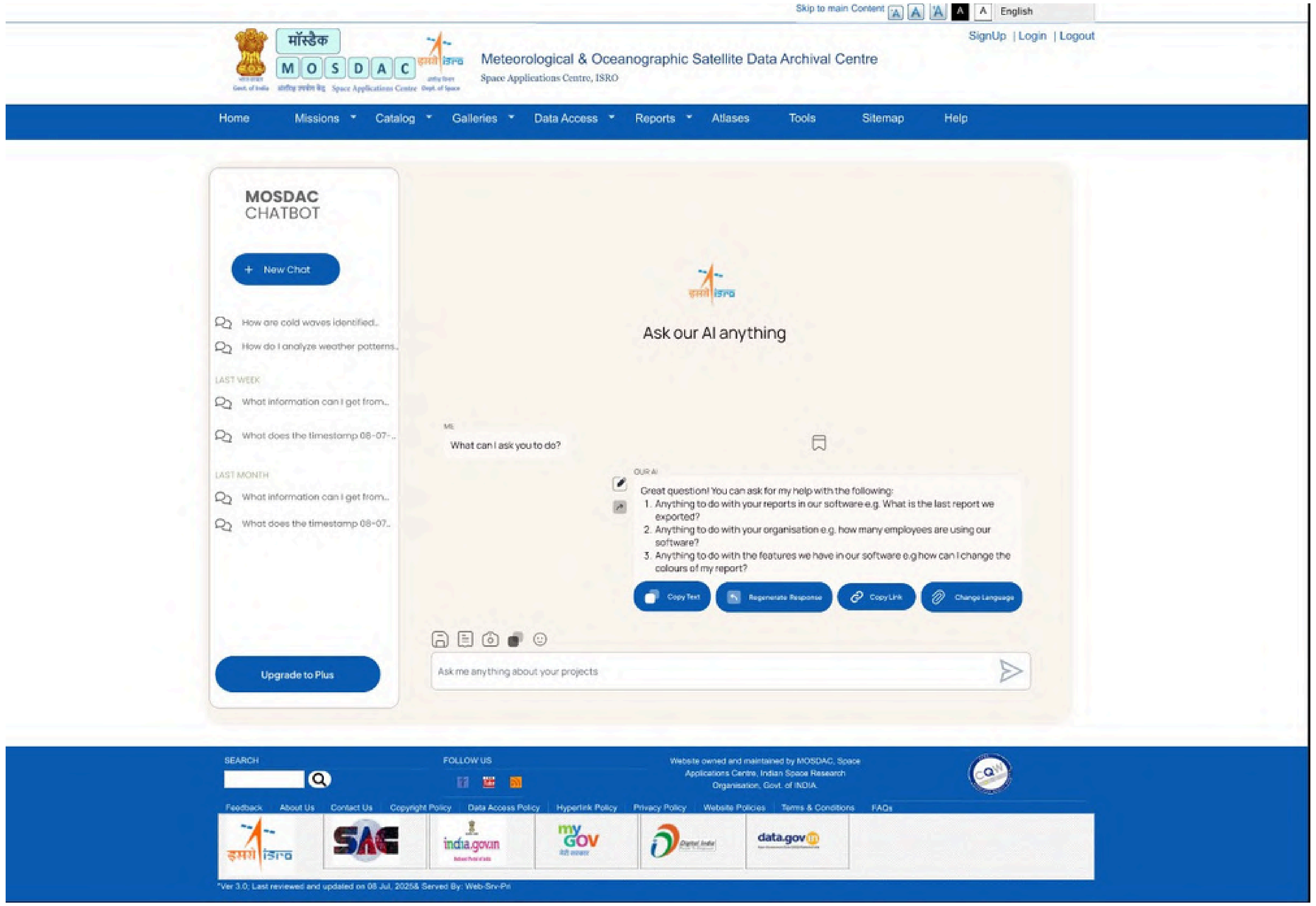


Wireframes/Mock diagrams of the proposed solution (optional)

Powered by H2S



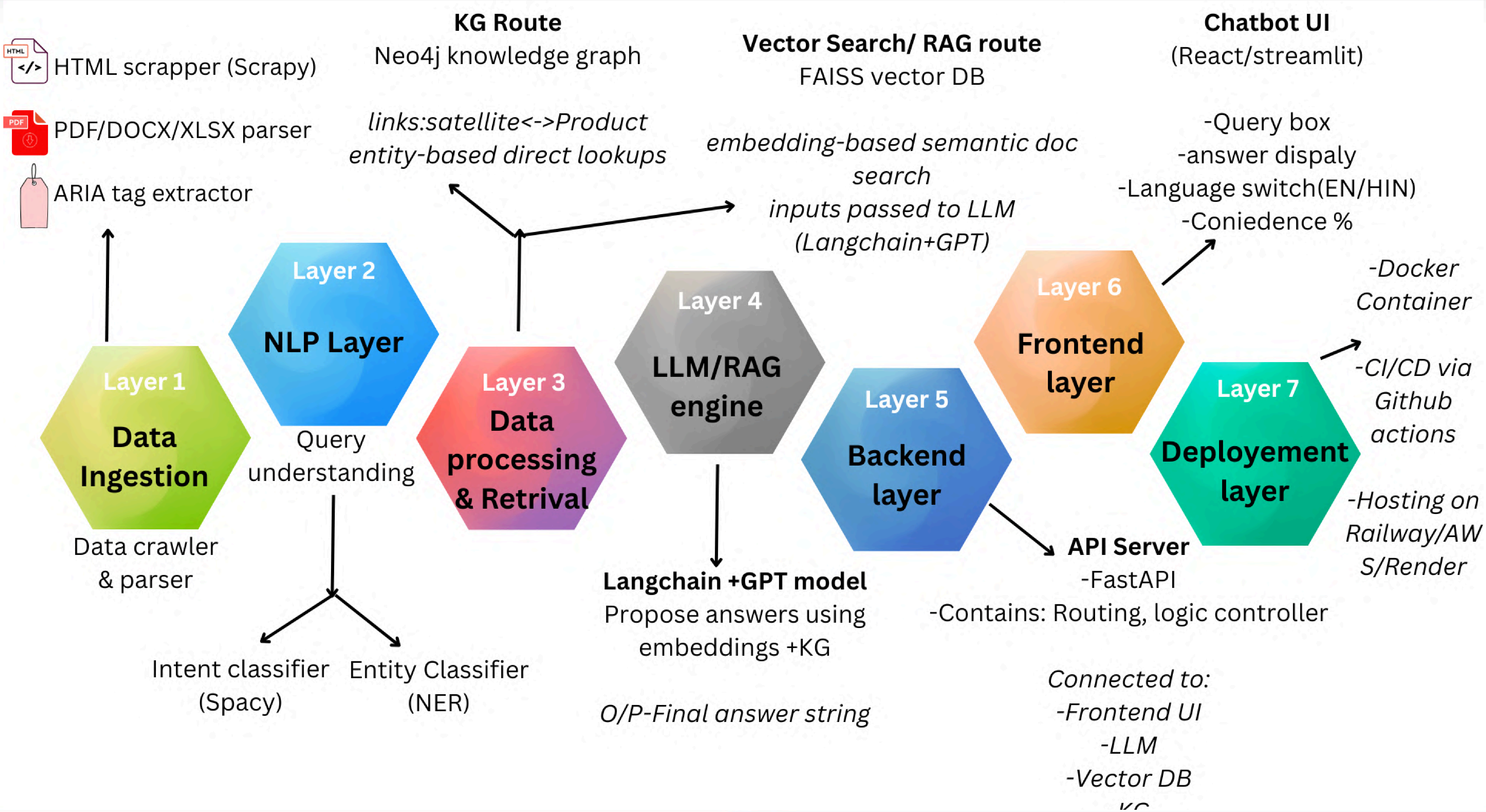
Chatbot Option at MOSDAC Portal



Overview of Chatbot

Architecture diagram of the proposed solution

Powered by



Technologies to be used in the solution:

Python — *Core language for backend logic, crawling, NLP, and integration.*

FAISS (Vector Search Engine) — *Enables fast semantic search across embedded documents.*

Scrapy + BeautifulSoup — *To crawl static and dynamic content (HTML, metadata, tables) from MOSDAC.*

LangChain + OpenAI/GPT — *For Retrieval-Augmented Generation (RAG) — combines context with generative AI.*

PyMuPDF, python-docx, openpyxl — *Parsing and extracting content from PDFs, DOCX, XLSX files.*

Streamlit / React.js — *Frontend chatbot interface (choose one based on your speed/UX needs).*

spaCy + NLTK — *For Named Entity Recognition (NER), intent detection, and tokenization.*

FastAPI — *Lightweight backend server to handle API requests and integrate modules.*

Neo4j (Knowledge Graph) — *To model entities like satellites, missions, and products with their relationships.*

Docker + GitHub Actions — *Containerization and automated CI/CD for scalable, modular deployment.*

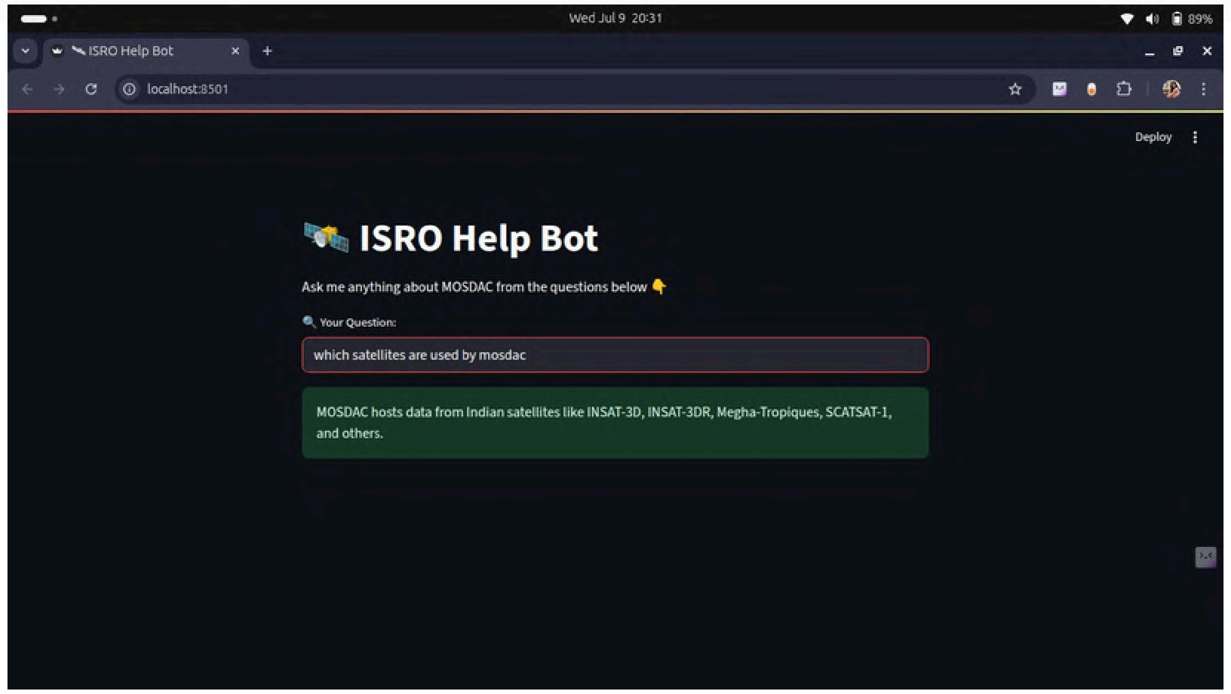
Estimated implementation cost:

Component	Tool/Service	Estimated Cost (INR/month)
Cloud Hosting	Railway / Render / AWS (free tier with upgrade)	₹2,000
Vector Search Engine	FAISS (local, open-source)	₹0
LLM API (for RAG)	OpenAI GPT-4 (light usage)	₹3,000–₹5,000
Database (KG + storage)	Neo4j Community Edition / SQLite	₹0
Frontend + Backend Hosting	Streamlit / FastAPI (on Render)	₹1,000
Miscellaneous	Domain, SSL, minor tools	₹1,000

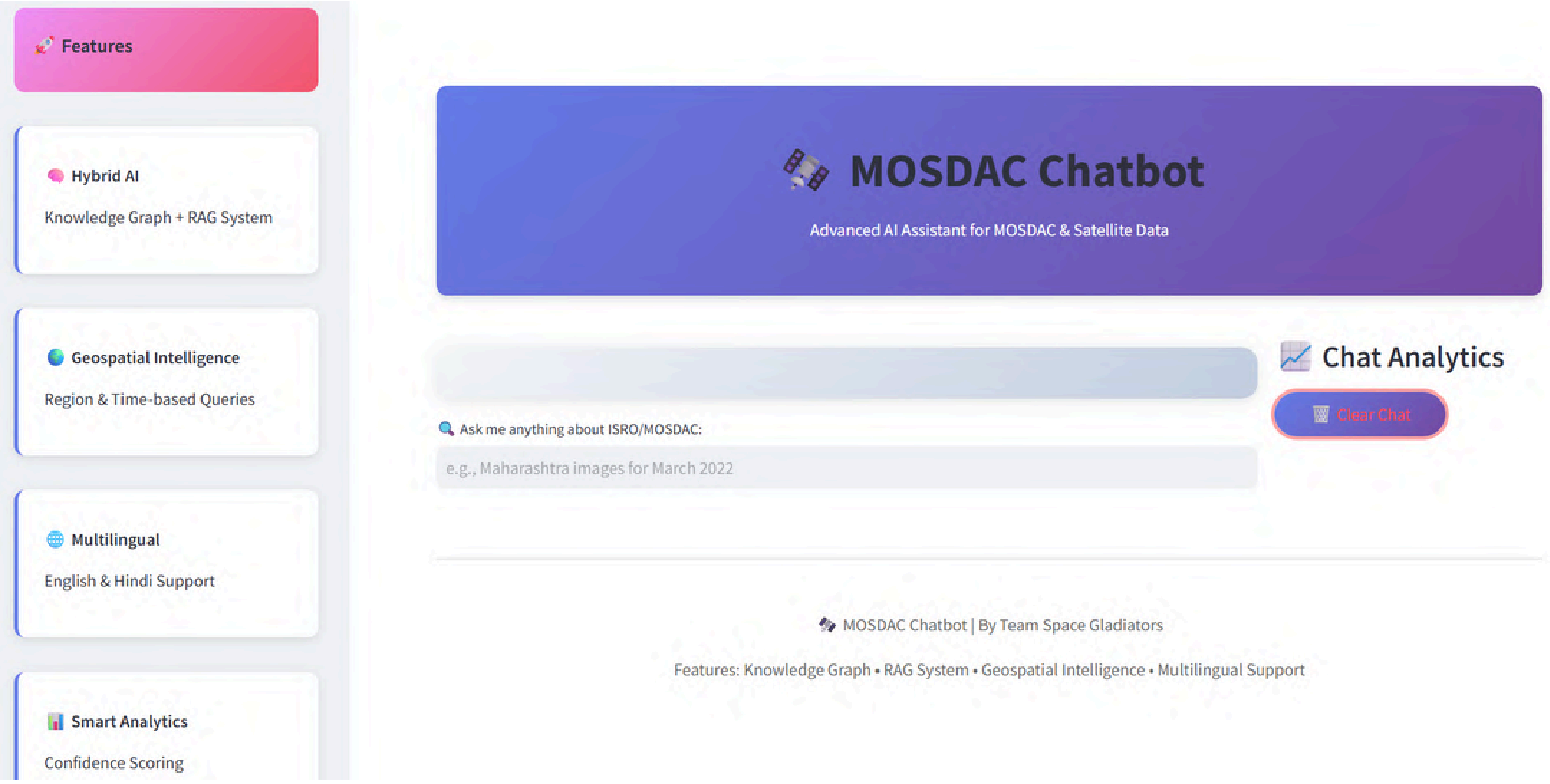
Total Estimated Cost (For MVP – 3 Months): ₹15,000 – ₹20,000

Prototype Screenshots:

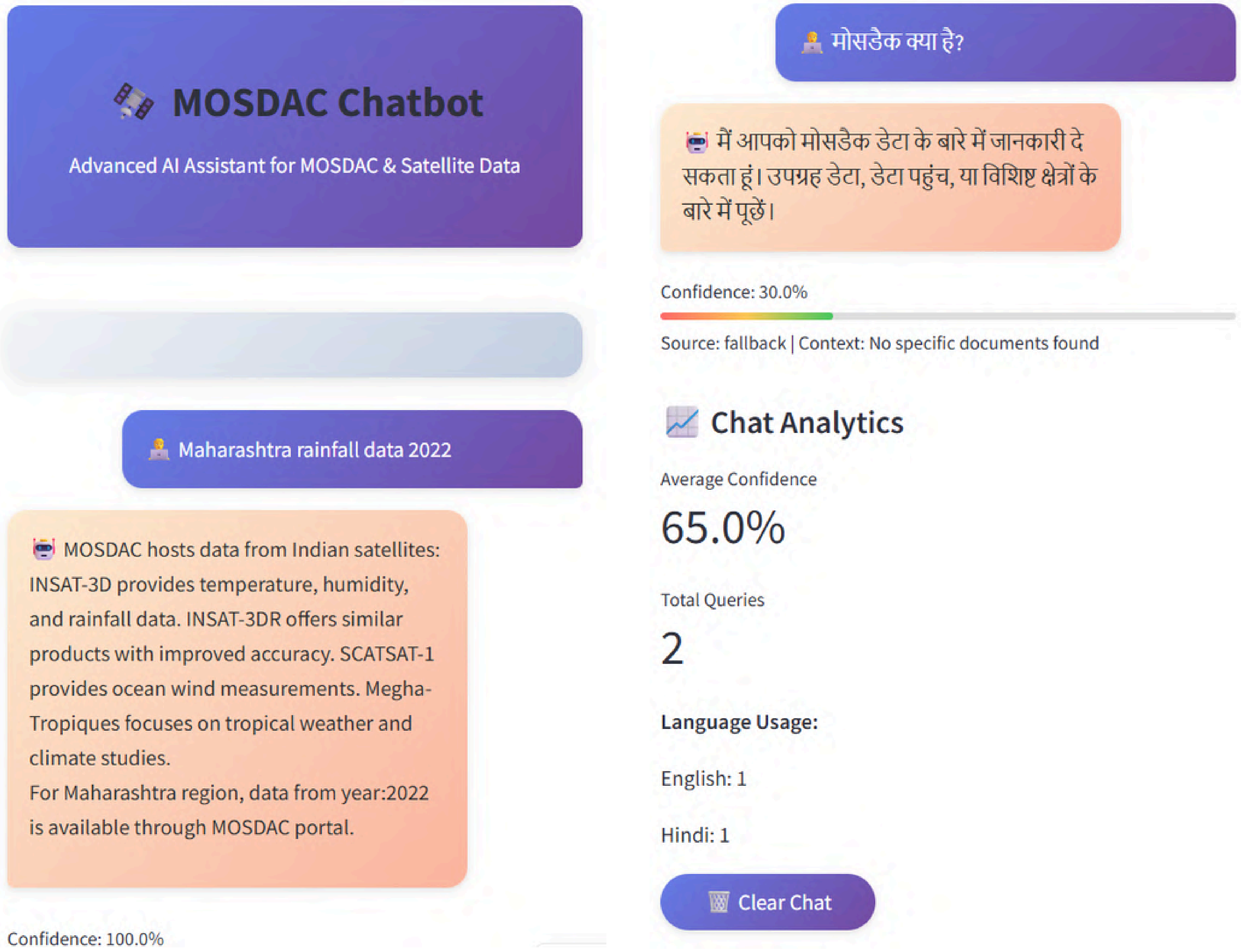
Powered by



Prototype 1 - MOSDAC Chatbot



Prototype 2 - MOSDAC Chatbot - Desktop Version (For Researchers)



Prototype 2 - MOSDAC Chatbot - Mobile Version (For Simple User Farmers)

Powered by H2S

BHARATIYA ANTARIKSH HACKATHON 2025

THANK YOU